

Exercise 1 - MMIOB

1) a) Sampling = Process of measuring an continuous-time analog signal at a uniform interval.

Quantization = A process to mapping a sampled amplitude value to the nearest value

b) $F(x) = \sin(2\pi f x + \varphi)$, $A = 10\text{ V}$, 5 bits quantization
 $2(\dots) = 20$

$$\text{minimum} = -10$$

$$\text{maximum} = 10$$

$$\text{maximum val swing} = 20\text{ V}$$

$$V_{pp} = 20 / 2^5 = 20 / 32 = 0.625\text{ V}$$

$$\text{Maximum quantization error} = 0.625 / 2 = 0.3125\text{ V}$$

$$c) F(x) = \sin(0.8\pi x) + \sin(1.5\pi x) + \sin(3\pi x)$$

$$F_1 = 0.8 / 2 = 0.4$$

$$F_2 = 1.5 / 2 = 0.75$$

$$F_3 = 3 / 2 = 1.5 \rightarrow F_{\text{max}}$$

$$\text{minimum sampling rate} = F_{\text{max}} \times 2 = 3$$

d) Aliasing is a corruption that happens when the rate of the sampling is slower than the nyquist rate, when the high frequency components that exceed the nyquist rate folds back to the lower frequency rate.

2a) Structured = managed by query language (DBMS)

Unstructured = managed by information Retrieval (Fuzzy matching)

Semi-Structured = Combines part of both (multimedia databases)

b) Structured = Information matching can be exact and fast

Unstructured = Slower and doesn't have exact matching

Semi-structured = Slowest for data retrieval and matching but most flexible

CD Rainbow Books

Book	Media	Additions
CD-DA	Continuous Audio	24 bit 79mins Audio
CD-ROM	File Storage and Error S.	Non Continuous Files
CD-i	Interactive Elements	Embedding of dif. Types
CD-M/R/RW	Dynamic Filestorage	Multi session
Video-CD	Mix of cont. and interactive	High fidelity video formats
E-CD	Cont. Audio w. File storage	Compatibility w. Multi S.
SuperAudioCD	Cont. Audio w. copy prot.	Policy Enforcements
DDCD	File Storage ...	new Storage Structure